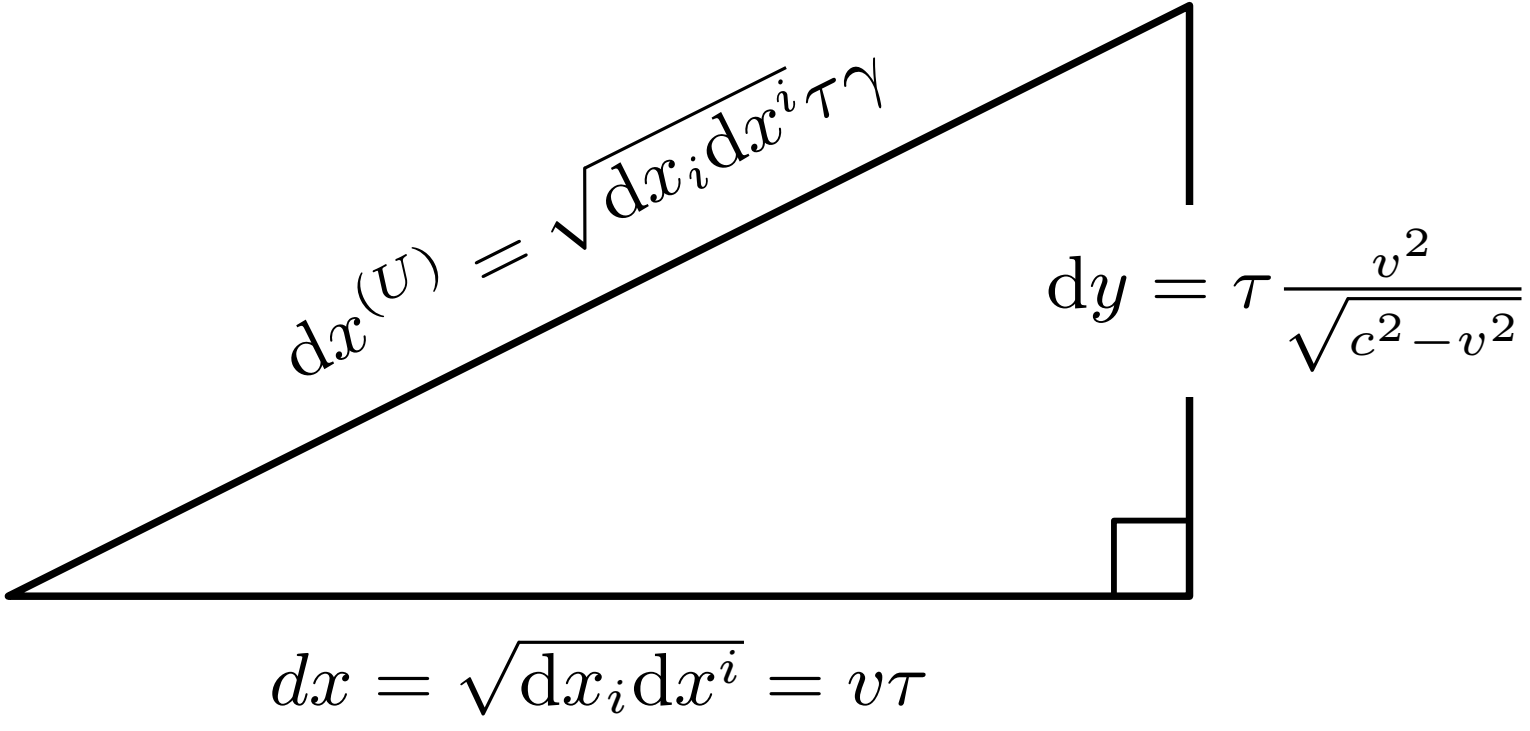

$\mathrm{d}x^{(\mathrm{U})} = \sqrt{\mathrm{d}x_i \mathrm{d}x^i}\,\tau\gamma$
$\mathrm{d}y = \tau \frac{v^2}{\sqrt{c^2 - v^2}}$
$dx = \sqrt{\mathrm{d}x_i \mathrm{d}x^i} = v\tau$